



Estimation of time-dependent laser heat flux distribution based on BPNN improved by multiple population genetic algorithm

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ABSTRACT

This study presents a method for estimating the time-dependent laser heat flux distribution of a nonlinear heat conduction system using a Back Propagation Neural Network improved by Multiple Population Genetic Algorithm (MPGA-BPNN) from transient temperature measurements. The primary focus of this research was on establishing a two-dimensional heat transfer model and training the MPGA-BPNN based on temperature distributions derived from finite volume simulations and any known heat flux. A series of numerical simulations were conducted to verify the feasibility of the proposed method, with corresponding experiments designed to test the method's efficacy. A comparative analysis was also conducted among BPNN, GA-BPNN and MPGA-BPNN methods. When the actual heat flux value is $4 \times 10^5 \text{ W/m}^2$, the relative errors for the BPNN, GA-BPNN, and MPGA-BPNN algorithms are 6.5%, 5.0%, and 1.0%, respectively. Furthermore, the MPGA-BPNN method has shown significant predictive capabilities in reliability and accuracy analyses across a range of cases. The study investigated the influence of network parameters and input temperatures on the heat flux outcomes using the MPGA-BPNN method. The heat flux of the laser on the sample surface was experimentally measured using this method, along with temperature distributions at 1.0–5.0 s obtained via a thermal imager.

1. Introduction

Laser heating, an efficient local heating technique, has a wide range of applications across various fields, including material processing [1–3], semiconductor manufacturing [4], the medical field [5], and scientific research [6,7]. The heat flux generated by laser heating plays a pivotal role in process control and optimization in these applications. Precise control of the heat flux by laser heating can improve semiconductor product quality, and increase production efficiency [8]. In the medical and precision processing fields, an understanding of the distribution of heat flux is crucial for the avoidance of thermal damage to sensitive areas [5]. Furthermore, heat flux data is of significant importance for the evaluation of the thermal properties and stability of materials [9,10], which in turn aids in material selection and application [11]. The acquisition of heat flux data is of paramount importance for the assurance of the efficiency, quality, and safety of laser applications.

As engineering technology advances, there is an increasing demand for heightened accuracy and rapid response times in heat flux measurements. Traditional transient heat flux meters necessitate calibration and judicious installation due to their sensitivity to experimental

conditions [12,13]. The precision of these meters is profoundly affected by factors such as physical dimensions, thermophysical properties, junction positioning, and inertia of the sensor materials [14]. Directly measuring heat flux in extreme thermal environments remains a formidable challenge. Given the complexities of direct measurement, heat flux inversion emerges as a viable approach [15–17]. Heat flux inversion entails reconstructing the heat flux distribution from surface temperature readings. In contrast, predictive methods aim to forecast future shifts in heat flux based on existing data and models. The advancement of these methodologies holds significant implications for enhancing system efficiency, ensuring equipment safety, and refining material design.

In recent years, there have been significant advances in the field of heat flux inversion and prediction methods. These developments encompass a range of techniques, including traditional analytical solution-based and numerical model-based approaches, as well as more sophisticated algorithms that leverage machine learning and deep learning techniques. A multitude of tools have been developed by researchers with the objective of enhancing the precision and efficiency of heat flux inversion. The predominant traditional inversion algorithms were the Conjugate Gradient Method (CGM) [18–22] and the Sequential

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Nomenclature		z	thickness direction, mm
A		<i>Greek symbols</i>	
A	heat flux distribution parameter 1	α	crossover probability
B	heat flux distribution parameter 2	γ	mutation probability
C	heat flux distribution parameter 3	Δ	random perturbation
C_p	sample specific heat, $\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$	ε	total emissivity
D	heat flux distribution parameter 4	ξ	random number
F	fitness function	θ	angle, $^\circ$
f	sigmoid activation function	ρ	sample density, $\text{kg}\cdot\text{m}^{-3}$
h	heat transfer coefficient, $\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$	σ	relative error
k	thermal conductivity, $\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$	τ	time, s
L	sample thickness, mm	$\Delta\tau$	time interval, s
p	selection probability	ϑ	threshold
Q	base heat flux value, $\text{W}\cdot\text{m}^{-2}$	<i>Acronyms</i>	
q	incident heat flux, $\text{W}\cdot\text{m}^{-2}$	BPNN	Back Propagation Neural Network
q_s	incident heat flux of prediction, $\text{W}\cdot\text{m}^{-2}$	CS	Constant Model
q_t	incident heat flux of model, $\text{W}\cdot\text{m}^{-2}$	FGS	Flattened Gaussian Model
R	sample diameter, mm	FVM	Finite Volume Method
R_l	heat flux diameter, mm	GS	Gaussian Model
r	radius direction, mm	MFGS	Modified Flattened Gaussian Model
T	sample temperature, K	MPGA	Multiple Population Genetic Algorithm
T_0	initial temperature, K	MSE	Mean Squared Error
T_f	ambient temperature, K	SGA	Sample Genetic Algorithm
v	input temperature		
w	synaptic weights of neuron		

Function Method (SFM) [23–25]. Subsequent iterations built upon these foundational models, leading to their refinement and perfection. Zhou et al. [18,20] employed the CGM, utilizing temperature and heat flux measurements from the back surface to ascertain the heat flux and temperature on the front (heated) surface of a three-dimensional (3D) object. Subsequently, they developed a nonlinear CGM scheme that incorporated radiation boundary conditions and temperature-dependent thermophysical properties. Kim et al. [23] and Wang et al. [24] applied the Sequential Function Specification Method (SFSM) to determine the spatial and temporal heat flux distributions for creep feed grinding and bond grinding, respectively. Besides, some ARX-based methods have always been employed in thermal engineering and other fields, as well as mathematics and automation [26,27]. Ríos-Moreno et al. [28] and Loussouarn et al. [29] demonstrated that the ARX model can be adopted to predict classroom indoor air temperature with high level of accuracy, as well as the temperature of ceramic parts in induction furnaces under vacuum conditions. In a further contribution to the field, Oliveira et al. [30] presented a novel inverse technique, named inverse ARX or IARX, which enables the estimation of inputs by direct calculation with identified parameters, based on the calibration step as a polynomial model. The distinction between IARX and ARX lies in the incorporation of future exogenous parameters, which were derived from the initial discrete form of the ARX model. In a numerical example utilizing IARX for a one-dimensional heat conduction simulation, the input heat flux was estimated with the consideration of discontinuities and measurement noises. The method demonstrated a faster calculation speed and a reduction in the required memory space. Additionally, some analytical solution-based methods have been studied for years [31,32]. Zhou et al. [33] proposed a new method based on Duhamel's theorem to predict the time-varying surface heat flux. This method can be implemented to solve multilayer, semi-infinite inverse heat conduction problems by establishing an auxiliary problem to obtain the transfer function numerically. Kant et al. [34] conducted an inversion of two-dimensional time-varying heat flux based on the exact solution of the conduction equation. They employed a three-dimensional sensitivity approach based on Duhamel's theorem to estimate transient, multidimensional heat flux distributions by measuring the temperature

distribution on the surface. In addition, a number of other inverse techniques have been employed in the field of heat conduction, including singular value decomposition [35], space marching method [36,37], genetic algorithm [38], particle swarm optimization [39], and digital filtering [40–42]. However, these methods are dependent on analytical solutions and numerical models. They lack universality, are only applicable to specific dimensions, or require stringent pre-conditions for validity. The accuracy and applicability of these techniques are often compromised when dealing with complex geometries or unknown boundary conditions. Furthermore, heat flux inversion is predominantly limited to single-point measurements, with measurement accuracy heavily reliant on the thermocouple.

The evolution of computing and storage capabilities has led to a significant advancement in machine learning intelligence algorithms for heat flux estimation. Notable examples include Gradient Boosting Decision Tree (GBTE) [43], Bayesian framework [44], Artificial Neural Networks (ANN) [45–47], and Back Propagation Neural Network (BPNN) [48,49]. These methods facilitate inversion without the necessity of an exact physical model by discerning intricate relationships between heat flux and temperature from training data. These methods demonstrate considerable potential in handling large datasets and capturing nonlinear patterns in data. The robustness and adaptability of these techniques, particularly their customizability for diverse optimization problems, render them optimal for tackling multi-objective optimization challenges. Nevertheless, these intelligent algorithms are not without limitations. Some researchers have proposed preliminary enhancements to these algorithms. For instance, Wu et al. [43] utilized GBDT machine learning algorithm to inverse the heat flux varying with time, with the relative error of 0.8% and 2.9% for the inversed results from the simulation and experiment. In addressing industrial heat conduction systems with complex geometries and online estimation requirements, Huang et al. [45] integrated ANN with an Adaptive Sequential Tikhonov Regularization (ASTR) method to address 3D nonlinear inverse problems. The ASTR method is well-suited to online processes due to its independence from future measurement data. Similarly, Wang et al. [46] proposed a new inverse estimation method of hot-wall heat flux, utilizing nonlinear artificial neural networks (ANN)

instead of an ideal explicit analytical physical model. The overall combined expanded uncertainty was found to be $\pm 9.4\%$ ($k = 2$) for the proposed inverse estimation of hot-wall heat flux. Furthermore, Wei et al. [50] addressed the issue of random connection weight settings during BPNN training, which can introduce errors. They utilized a genetic algorithm to optimize the internal weight and threshold of the BPNN, aiming for global optimization. Nevertheless, genetic algorithms continue to be constrained by inherent limitations, including the necessity to select an initial population and to determine operator parameters.

In this study, we introduce a method that leverages the BPNN enhanced by multiple population genetic algorithm (MPGA) to estimate transient heat flux in non-linear heat conduction systems. The BPNN, with a fixed input length, is trained using training data derived from Finite Volume Method (FVM) simulations. The initial weights and thresholds for the BPNN are determined through MPGA optimization. The MPGA effectively balances global and local searches by co-evolving populations with varied control parameters. This approach addresses the limitations of GA's reliance on initial population selection and operator parameter settings, and it transcends the conventional single-population evolution paradigm of traditional genetic algorithms. To validate the computational accuracy and efficiency of our proposed MPGA-BPNN algorithm, we conducted numerical tests and compared it with several other methods. Furthermore, we analyzed the effects of network parameters and input measurement noise on the heat flux outcomes. Additionally, experimental measurements of heat flux distribution on the sample surface were completed.

2. Experiment and method

2.1. Experiment setup

The schematic diagram of the self-designed apparatus for the measurement of laser heat flux distribution is shown in Fig. 1. The experiment utilizes a high-power continuous laser with a single wavelength of $0.92 \mu\text{m}$, the power of which is regulated by software up to 1000 W. The laser power employed in the experiments was 100 W, with a laser spot diameter of 20 mm. The laser heat flux exhibits a Gaussian distribution, which can be converted into a flattened beam distribution using a laser beam shaper. Nevertheless, the post-conversion value and distribution of the laser heat flux remain undetermined. The target material may comprise homogeneous materials that exhibit robust thermal shock resistance due to the high heat flux density of the laser under

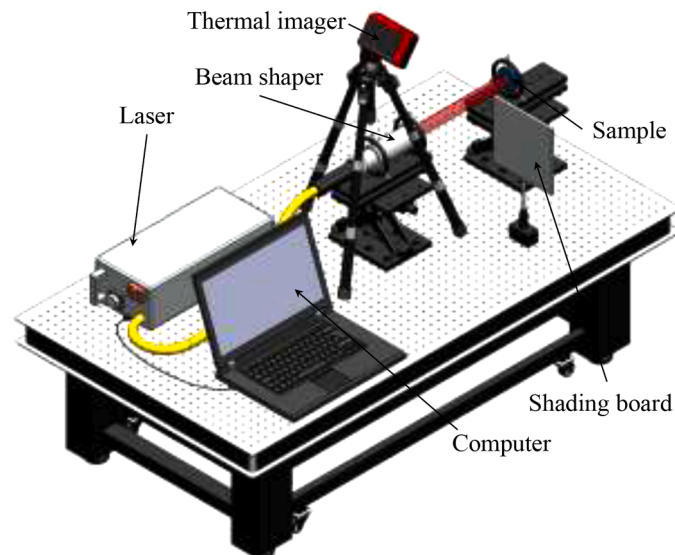


Fig. 1. Schematics of the laser heat flux distribution measurement apparatus.

examination. Silicon carbide ceramic is opaque to the laser wavelength and offers benefits such as stable composition and cost-effectiveness.

Table 1 provides a summary of the detailed geometric parameters and key thermophysical properties of the silicon carbide ceramic material. It should be noted that the specific heat capacity, thermal conductivity, and total emissivity are measured by the experimental apparatus in our research group [51,52]. The emissivity represents an integrated mean value across the wavelength range from $7 \mu\text{m}$ to $14 \mu\text{m}$. Given that the experiment measurements were conducted indoors, a constant nature convection heat transfer coefficient of $10.0 \text{ W}/(\text{m}^2\cdot\text{K})$ was selected to simplify the calculation model. The initial and ambient temperatures were both 298.15 K. The infrared thermal imager (Fotric 287) employed in the experiment has a resolution of 512×384 pixels within the $7\text{--}14 \mu\text{m}$ waveband and captures thermal images at an angle of 30° . It should be noted that the shading board is moved once the laser power has stabilized. Subsequently, the transient temperature distribution on the front surface of the sample is captured by the infrared thermal imager for identification of heat flux distribution.

2.2. Method formulation

2.2.1. Mathematical model

Fig. 2 depicts the physical model for the direct problem in a two-dimensional cylindrical system, as investigated in this study. As illustrated in Fig. 2(a), the diameter and thickness of the cylinder are denoted as R and L , respectively, while the diameter of the laser beam is represented by R_1 . Given that the incident heat flux distribution is symmetrically oriented about the z -axis and the thermal conduction process along the axis direction of the homogeneous material, the temperature field within the sample is also symmetrically distributed about the z -axis, as illustrated in Fig. 2(b). Consequently, the temperature distribution remains unaffected by the circumference angle θ and is instead a function of radius r , thickness z , and time τ . The initial temperature of the sample is T_0 , and the ambient temperature is T_f . Once the experimental duration exceeds $\tau > 0$, an unknown incident heat flux $q(r, \tau)$ of the laser, with its distribution varying with both radius and time, is applied to the front surface where $z = L$ and $r \leq R_1$. On the surfaces of the sample and within the region of the heat flux's diameter R_1 , boundary conditions at boundaries $z = 0$, $z = L$, and $r = R$ are defined as convective heat transfer boundary conditions and radiation boundary conditions.

A mathematical model for a transient heat conduction problem in a two-dimensional finite length rectangle with homogeneous material is presented. This model employs Cartesian coordinates to delineate the temperature profile of the sample through non-Fourier heat conduction, as outlined in Eq. (1).

$$\rho C_p(T) \frac{\partial T}{\partial \tau} = \frac{\partial}{\partial x} \left(k_r(T) \frac{\partial T}{\partial r} \right) + \frac{\partial}{\partial z} \left(k_z(T) \frac{\partial T}{\partial z} \right), 0 \leq r \leq R, 0 \leq z \leq L \quad (1)$$

where ρ , C_p , and k are density, specific heat, and thermal conductivity of material, respectively, τ is the time, T is the temperature of material. As described above, the boundary conditions and initial conditions can be

Table 1
Parameters of the materials and experimental conditions in the system.

Parameter	Value
Material	Silicon carbide ceramic
Geometrical parameter $2R \times L$ (mm)	$\Phi 30 \times 5$
Density ρ (kg/m^3)	3210
Specific heat capacity C_p ($\text{J}/(\text{kg}\cdot\text{K})$)	$735 - 8.21 \times 10^{-2} \cdot T$ (300~1000 K)
Thermal conductivity k ($\text{W}/(\text{m}\cdot\text{K})$)	$83.8 - 1.63 \times 10^{-4} \cdot T$ (300~1000 K)
Emissivity ϵ	0.63
Convective heat transfer coefficient h ($\text{W}/(\text{m}^2\cdot\text{K})$)	10.0
Initial temperature T_0 (K)	300.0
Ambient temperature T_f (K)	300.0

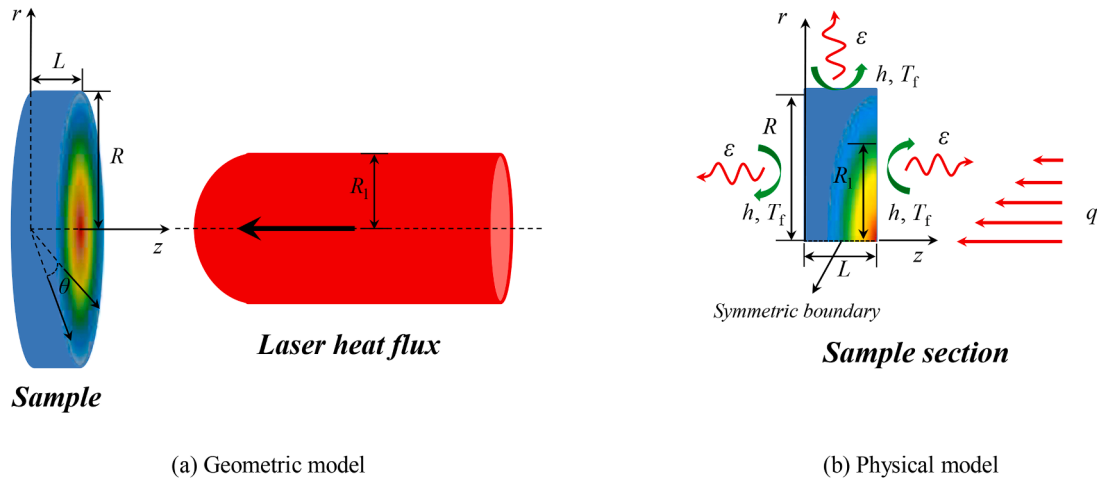


Fig. 2. Schematic of transient heat transfer model in a 2D cylindrical system.

shown as Eqs. (2-6),

Boundary conditions:

$$\begin{aligned} -k_z(T) \frac{\partial T}{\partial z} &= \varepsilon q(r, \tau) - \varepsilon \sigma (T^4 - T_f^4) - h(T - T_f), \quad 0 \leq r \leq R_1, \quad z = L \\ -k_z(T) \frac{\partial T}{\partial z} &= -\varepsilon \sigma (T^4 - T_f^4) - h(T - T_f), \quad R_1 < r \leq R, \quad z = L \end{aligned} \quad (2)$$

$$-k_z(T) \frac{\partial T}{\partial z} = -\varepsilon \sigma (T^4 - T_f^4) - h(T - T_f), \quad 0 \leq r \leq R, \quad z = 0 \quad (3)$$

$$-k_r(T) \frac{\partial T}{\partial r} = 0, \quad r = 0, \quad 0 \leq z \leq L \quad (4)$$

$$-k_r(T) \frac{\partial T}{\partial r} = -\varepsilon \sigma (T^4 - T_f^4) - h(T - T_f), \quad r = R, \quad 0 \leq z \leq L \quad (5)$$

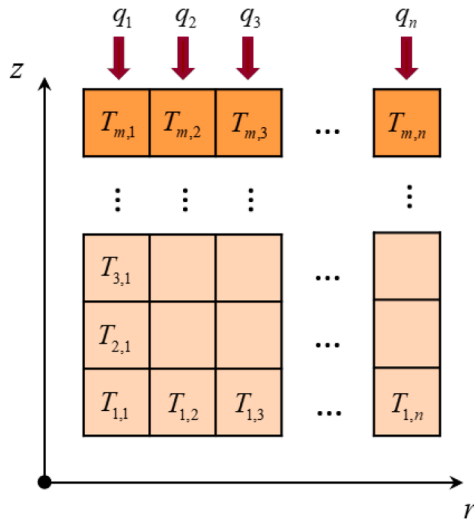
Initial conditions:

$$T = T_0, \quad \tau = 0 \quad (6)$$

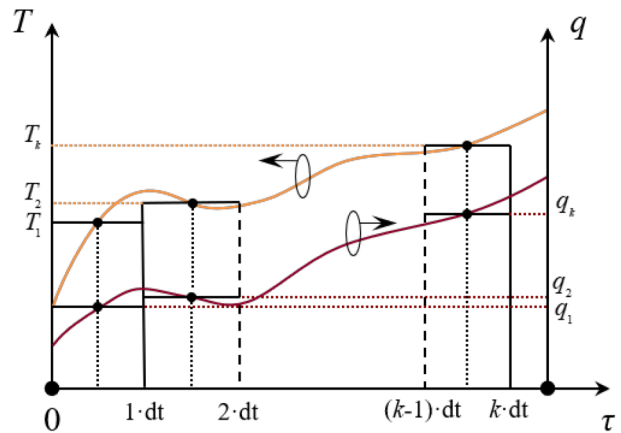
where h is the convective heat transfer coefficient between material surface and air, and the boundaries are diffuse surfaces with the same

emissivity (ε), respectively. It should be noted that the heat transfer modeling of the material is constrained by the assumption that the heat flux does not penetrate the material. The aforementioned Eqs. (1-6) determine the interior temperature $T(r, z, \tau)$ of the material, contingent upon specified thermophysical properties and initial and boundary conditions. The direct problem can be addressed using Finite Volume Method (FVM) simulation.

Then, the sample section is discretized in both space and time, as illustrated in Fig. 3. Fig. 3(a) depicts the spatial discretization of the sample section across $m \cdot n$ temperature elements, where m denotes the number of elements in the z -direction and n denotes the number of elements in the r -direction. The local index for an individual element is denoted as (i, j) . In general, temperature observers such as thermocouples are typically positioned on the upper surface, with elements ranging from $(m, 1)$ to (m, n) being used instead. Alternatively, they can be placed within the sample section itself. In this part, the numbers of m and n need to depend on the pixels of infrared thermal imager. Furthermore, the temperature of each element (i, j) is temporally discretized, as depicted in Fig. 3(b). Temperatures at times $\tau = 0.5 \cdot dt, 1.5 \cdot dt, \dots, (k-0.5) \cdot dt$ represent the average temperatures within ranges $\Delta\tau = 0 \sim 1 \cdot dt, 1 \sim 2 \cdot dt, \dots, (k-1) \sim k \cdot dt$. Similarly, the heat flux on the



(a) In space



(b) In time

Fig. 3. Discretization in space and time of sample section surface for temperatures and heat fluxes.

upper surface is also discretized analogously to the temperature both spatially and temporally [34].

2.2.2. MPGA-BPNN algorithm

Fig. 4 shows the novel methodology proposed in this work, termed as named Back Propagation Neural Network improved by Multiple Population Genetic Algorithm (MPGA-BPNN). This method accurately predicts heat flux using directly measured parameters such as temperature, and is based on the Back Propagation Neural Network (BPNN) with input, hidden, and output layers. The dimensions of both the input and output layers of the prediction network can be tailored based on influential factors and outcomes within the Inverse Heat Conduction Problem (IHCP). The number of units in the hidden layer is crucial and directly correlates with predictive accuracy. As depicted in Fig. 5, a three-layer BP neural network is employed to estimate heat flux, with input and output layer dimensions set at m and 4 respectively. It should be noted that m temperatures from the upper surface are utilized for inputs, while four parameters are used for outputs to represent the distribution of heat flux on the upper surface. Each neuron in the BP neural network exhibits a non-linear output, utilizing a sigmoid activation function as Eq. (7).

$$f(x) = \frac{1}{1 + e^x} \quad (7)$$

The neuron can be described by Eq. (8) in the mathematical terms:

$$\text{output} = f\left(\sum_{i=1}^n w_i v_i - \vartheta\right) \quad (8)$$

where $v_1, v_2, v_3, \dots, v_n$ represent the input temperatures, $w_1, w_2, w_3, \dots, w_n$ represent the synaptic weights of neuron, and ϑ represents the threshold. Utilizing the error-correction learning rule, adjustments to the weights and thresholds of the neural network are made via backpropagation. This process aims to minimize the cumulative sum of squared errors within the network.

The Simple Genetic Algorithm (SGA) can be employed to optimize the internal weight threshold of the Back Propagation Neural Network (BPNN), thereby preventing it from falling into a local extreme point. This enhanced BPNN, is also known as the Genetic Neural Network (GNN) or the BP Neural Network Improved by Genetic Algorithm (GA-

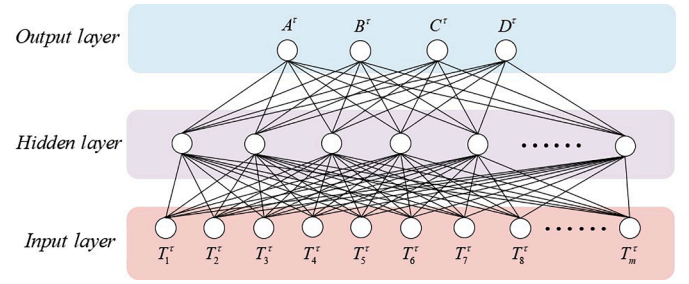


Fig. 5. Structure of the three-layer BP neural network.

BPNN). The GNN proposed by Wei et al. [50] employs six steps to achieve global optimal results: weight and threshold coding, initial population generation, fitness function calculation, selection operation, crossover operation, and mutation operation. These operational steps are defined in detail in reference [50]. The input temperatures are transformed into genetic genes in the chromosome. The chromosome is then randomly generated to form an initial population. The fitness function is defined as the half error of the square average between the predicted temperature T_i' and target temperature T_i , as defined in Eq. (9),

$$F_i = \frac{\sum_{i=1}^n (T_i - T_i')^2}{2n} \quad (9)$$

where F_i represents the fitness of the i th individual and n represents the number of input layer neurons. Subsequently, the selection operation employs a roulette wheel selection methodology to identify new individuals. The probability of individual selection p_i is calculated by Eq. (10).

$$p_i = \frac{F_i}{\sum_{i=1}^n F_i} \quad (10)$$

The crossover operation adopts an arithmetic crossover method, wherein the linear combination of two individuals generates two new

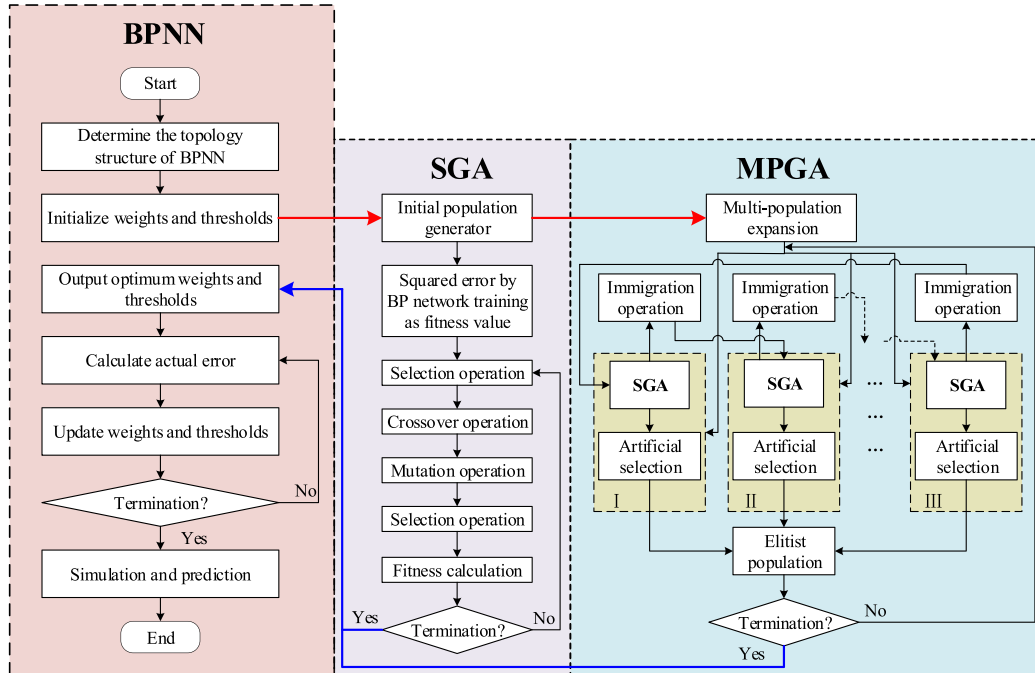


Fig. 4. Flow diagram of BP neural network improve by multiple population genetic algorithm.

individuals based on a crossover probability α , as shown in Eq. (11),

$$\begin{aligned} w_A' &= \alpha w_B^r + (1 - \alpha) w_A^r \\ w_B' &= \alpha w_A^r + (1 - \alpha) w_B^r \end{aligned} \quad (11)$$

where w_A^r and w_B^r represent the weights of neuron before crossover operation, w_A' and w_B' represent the weights of neuron after crossover operation. To circumvent the issue of genetic algorithms becoming ensnared in local optima during the optimization process, it is imperative to execute mutation operations on individuals throughout the search procedure. This can be achieved through a non-uniform mutation approach that involves applying a random perturbation Δ to the original gene value. The perturbed outcome is then considered as the mutated new gene value.

$$\begin{aligned} w_\tau' &= w_\tau + \Delta(w_\tau^{\max} - w_\tau), \text{ if } \xi_1 < \gamma \\ w_\tau' &= w_\tau - \Delta(w_\tau - U_\tau^{\min}), \text{ if } \xi_1 > \gamma \end{aligned} \quad (12)$$

w_τ and w_τ' represent the weights of neuron before and after mutation operation, $w_\tau^{\min}$ and $w_\tau^{\max}$ represent the minimum and maximum values of weights of neuron, ξ_1 is a random number between 0 and 1, and γ is the mutation probability. The random perturbation Δ is related to the number of iterations and can be given as:

$$\Delta = \left(\xi_2 \times \left(1 - \frac{N_{\text{now}}}{N_{\text{max}}} \right) \right)^2 \quad (13)$$

where N_{now} and N_{max} are the numbers of current and maximum iterations, ξ_2 is a random number between 0 and 1. However, the simple genetic algorithm exhibits a tendency towards premature convergence due to several factors. These include the high fitness of certain individuals, an excessive reliance on crossover and mutation probabilities, a limited group size, and insufficient iterations.

The Multiple population genetic algorithm (MPGA) is introduced to compensate for the deficiencies of the Single Genetic Algorithm (SGA). The computation process based on a back propagation neural network, enhanced by multiple population genetic algorithm (MPGA-BPNN), has been shown in Fig. 4. The multiple population genetic algorithm (MPGA) is constructed based on the SGA and several main produces are introduced as follows:

- (1) The proposed algorithm overcomes the limitations of Single Genetic Algorithm (SGA), which is traditionally reliant on a single population for genetic evolution. In contrast, it enables co-evolution across multiple populations with distinct control parameters, thereby achieving a harmonious balance between global and local search. This was substantiated through a series of simulation experiments, in which the following parameters were set: the number of populations was 12, the number of individuals was 100, the selection probability ranged from 0.7 to 0.9, the crossover probability was between 0.3 and 0.5, and the mutation probability was from 0.001 to 0.05.
- (2) Populations from various sources are interconnected with immigration operators, facilitating the co-evolution of multiple populations. The immigration operator serves as a mechanism to replace the least effective individual within the target population with the most efficient one from the source population. The attainment of this optimal solution represents the cumulative outcome of the co-evolutionary process across multiple populations.

$$w_\tau^l(i+1) = w_\tau^m(i) \quad (14)$$

$w_\tau^l(i+1)$ is the least effective individual within the target population $i+1$, and $w_\tau^m(i)$ is the most efficient one from the source population i .

- (3) The process of artificial selection allows for the preservation of elite populations from each evolutionary generation of various groups are preserved. The criteria for the selection of elite population is to identify the optimal individual within each population. In other words, the optimal individual is the one in the population that best fits the fitness function. This obviates the necessity for selection, crossover, mutation, or other operations to protect the optimal individual from destruction or loss throughout the evolutionary process. The minimum preservation algebra of the optimal individual serves as the foundation for assessing the convergence of the algorithm. This approach is more rational than the maximum genetic algebra, which enhances the optimization performance of genetic algorithms.

3. Numerical results and discussion

In numerical modeling, a continuous laser is used due to its continuously and stably output. It is predicted that the incident heat flux distribution of the laser will be Gaussian, and that the laser output end will be processed using a collimating mirror. Consequently, the distribution of incident heat flux varies with radius r and time t , as illustrated in Eq. (15). The temperatures calculated over the initial five seconds are considered as the measured temperatures for the training data.

$$q(r, \tau) = Q \cdot \exp \left(-A \times \left(\frac{r}{C} \right)^2 \right) \sum_{k=0}^N \frac{1}{k!} \left(B \times \left(\frac{r}{C} \right)^2 \right)^k \cdot D \quad (15)$$

$$0 \leq r \leq 10, 0 \leq \tau \leq 5, N = 2$$

The distribution of heat flux, as described by Eq. (15), is characterized by four parameters. The parameters A , B , C , and D are derived from the Constant Model (CS), Gaussian Model (GS), Flattened Gaussian Model (FGS), and Modified Flattened Gaussian Model (MFGS), respectively. The parameter ranges for these models are detailed in Table 2, with a base heat flux value of Q set at $2.0 \times 10^5 \text{ W/m}^2$. It should be noted that the diameter of the heat flux beam is 20 mm, and only the initial five seconds are considered. Consequently, the independent variable ranges for r and τ are 0–10 and 0–5, respectively. Table 3 presents four distinct distribution models of heat flux at the time of 1.0 s, which serve to validate the precision and adaptability of the MPGA-BPNN algorithm.

3.1. Reliability analysis of MPGA-BPNN algorithm

The direct model and the proposed MPGA-BPNN algorithm were utilized to analyze the variation of heat flux with distance from the center across different distributions and times, as depicted in Figs. 6 and 7. Fig. 6 shows the variations of simulated temperature and heat flux with the distance from center for different heat flux distribution models at 1.0 s. It can be observed in Fig. 6(a) that the simulated outcomes of the two models of CS and FGS are relatively similar, potentially due to their comparable heat flux distributions. Fig. 6(b) shows a comparison between the model q_t and the predictions of the MPGA-BPNN q_s for four distinct heat flux distributions. In the case of the proposed models, the heat flux q is distributed uniformly along the sample's circumference of the sample and varies with the sample radius x within the range of 0 to $4 \times 10^5 \text{ W/m}^2$. Similarly, Fig. 7 shows the variations of simulated

Table 2
Parameter values for the distribution of heat flux in Eq. (15).

Parameter	Value
A	0.0, 5.0, 1.0, ..., 5.0
B	0.0, 25.0, 0.5, ..., 3.0
C	5.0, 6.0, 7.0, ..., 15.0
D	0.5, 0.75, 1.0, ..., 2.5
$Q \text{ (W/m}^2\text{)}$	2.0×10^5

Table 3

Different distribution models of heat flux at 1.0 s.

Distribution models	$q(r, 1.0)$
Constant Model (CS)	$q(r, 1.0) = Q \times 2.0, 0 \leq r \leq 10$
Gaussian Model (GS)	$q(r, 1.0) = Q \cdot \exp\left(-2 \times \left(\frac{r}{6}\right)^2\right) \times 2.0, 0 \leq r \leq 10$
Flattened Gaussian Model (FGS)	$q(r, 1.0) = Q \cdot \exp\left(-\left(\frac{r}{10}\right)^2\right) \sum_{k=0}^N \frac{1}{k!} \left(\left(\frac{r}{10}\right)^2\right)^k \times 2.0, 0 \leq r \leq 10, N = 2$
Modified Flattened Gaussian Model (MFGS)	$q(r, 1.0) = Q \cdot \exp\left(-3.0 \times \left(\frac{r}{9}\right)^2\right) \sum_{k=0}^N \frac{1}{k!} \left(3.0 \times \left(\frac{r}{9}\right)^2\right)^k \times 2.0, 0 \leq r \leq 10, N = 2$

temperature and heat flux with the distance from center for the MFGS model at different times. In Fig. 7(a), the temperatures at different positions on the sample surface increase over time. Fig. 7(b) illustrates a comparison between the model q_i and the predictions of the MPGA-BPNN q_s for the fourth heat flux distribution MFGS at 1.0 s, 2.0 s, and

3.0 s. The results demonstrate a generally satisfactory agreement is observed across different distribution models and times. Tables 4 and 5 present the values of four distribution parameters for various heat flux distributions and times. The maximum relative error value for the four parameters is 6%, and the maximum deviation of heat flux is less than $1.0 \times 10^4 \text{ W/m}^2$. It can therefore be inferred that the MPGA-BPNN algorithm is highly suitable for measuring transient heat flux.

3.2. Influence of network parameters and input temperatures

This section presents an analysis of the influence of network parameters and input temperatures on heat flux predictions based on the MFGS model. Fig. 8 depicts the variation in Mean Squared Error (MSE) for MPGA-BPNN as a function of population size N_p and hidden layer neuron count N_h . As both population size and hidden layer neuron count increase, there is a gradual decrease in the MSE value of the MPGA-BPNN algorithm. However, when the population size reaches 10, further increases in population size have a negligible impact on improving the accuracy and flexibility of the MPGA-BPNN algorithm.

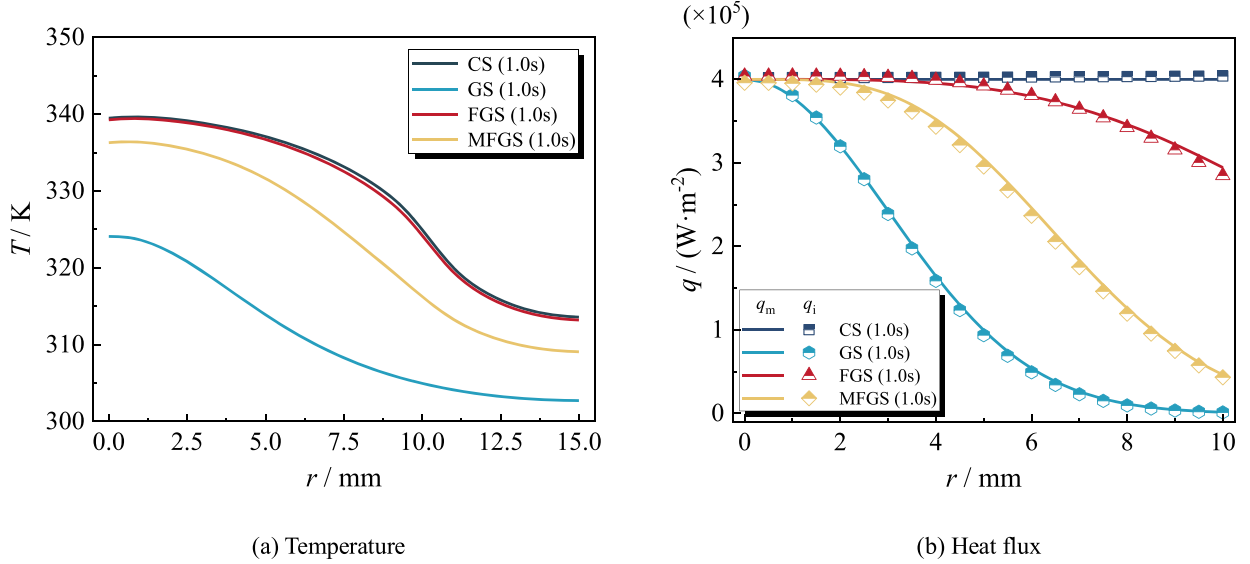
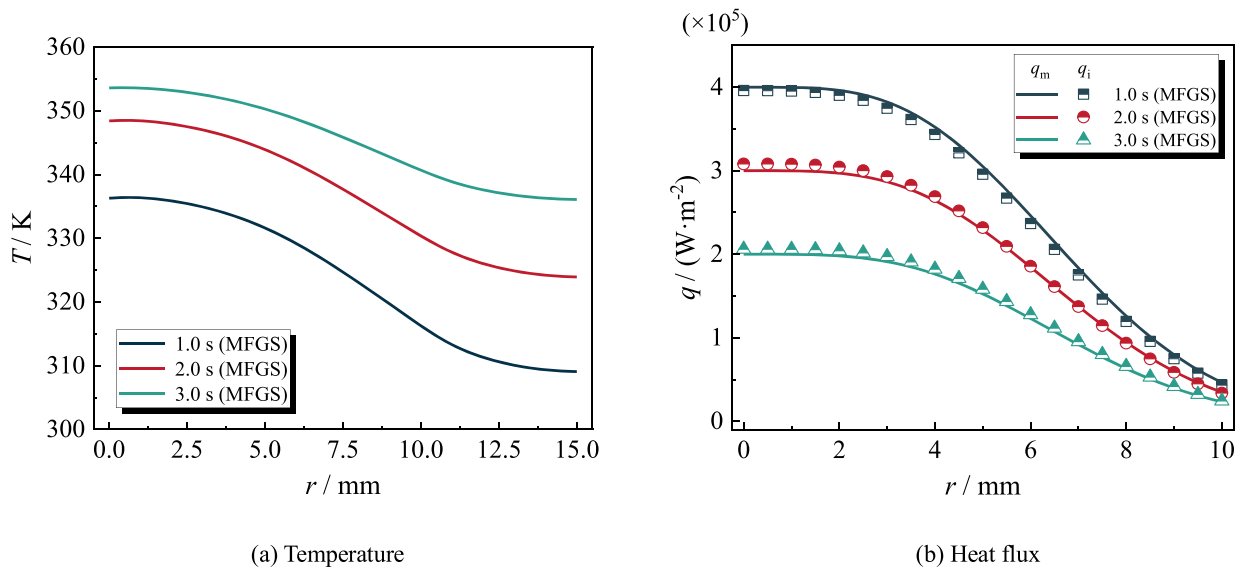
**Fig. 6.** Variations of surface temperature and heat flux with the distance from center for different heat flux distributions.**Fig. 7.** Variations of surface temperature and heat flux with the distance from center for different times.

Table 4

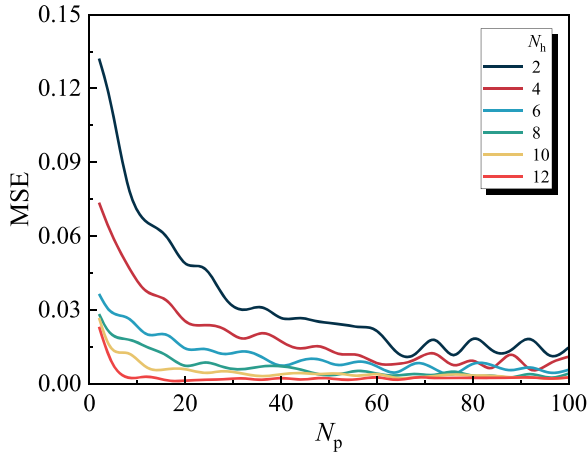
Compare distribution parameters from numerical simulation and MPGA-BPNN for different models.

Different models	A			B			C			D		
	q_t	q_s	σ	q_t	q_s	σ	q_t	q_s	σ	q_t	q_s	σ
CS (1.0 s)	0	0.01	—	0	0.03	—	—	—	—	2.00	2.01	0.5%
GS (1.0 s)	2.00	1.97	-1.50%	0	0.06	—	6.00	5.72	-4.7%	2.00	2.01	0.5%
FGS (1.0 s)	1.00	1.05	5.0%	1.00	1.06	6.0%	10.00	9.80	-2.0%	2.00	2.02	1.0%
MFGS (1.0 s)	3.00	2.91	-3.0%	3.00	2.83	-5.7%	9.00	8.81	-2.1%	2.00	1.98	-1.0%

Table 5

Compare distribution parameters from numerical simulation and MPGA-BPNN for different times.

Different times	A			B			C			D		
	q_t	q_s	σ	q_t	q_s	σ	q_t	q_s	σ	q_t	q_s	σ
1.0 s (MFGS)	3.00	2.91	-3.0%	3.00	2.83	-5.7%	9.00	8.81	-2.1%	2.00	1.98	-1.0%
2.0 s (MFGS)	3.00	2.95	-1.7%	3.00	2.91	-3.0%	9.00	8.85	-1.7%	1.50	1.54	2.7%
3.0 s (MFGS)	3.00	2.94	-2.0%	3.00	2.96	-1.3%	9.00	8.93	-0.8%	1.00	1.03	3.0%

**Fig. 8.** Variation of MSE of MPGA-BPNN for with number of populations N_p and hidden layer neurons N_h .

When the population size is fixed, the MSE initially decreases sharply before fluctuating within a certain range as the number of neurons increases. It can be observed that with a population size of 12 and 20 neurons in the hidden layer, the relative error is minimized, indicating that there is no need to increase the population size to overload neural networks.

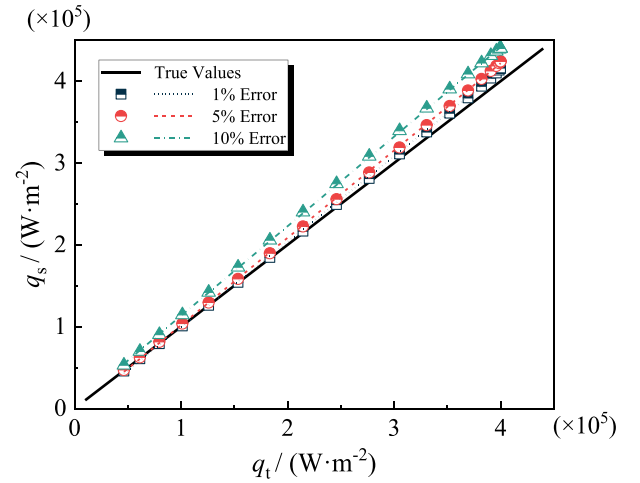
To consider the impact of various uncontrollable factors on the measurement results during the experimental process, random errors can be added to the material temperature T_{exact} obtained from the heat flow model to simulate the temperature T_{exp} measured under experimental conditions:

$$T_{\text{exp}} = T_{\text{exact}} + \varphi\zeta \quad (16)$$

Here, ζ is a normal distribution random variable with zero mean and unit standard deviation. The standard deviations of measured temperature, φ for a $\gamma\%$ measured error at 99% confidence, are determined as:

$$\varphi = \frac{T_{\text{exact}} \times \gamma\%}{2.576} \quad (17)$$

where the value of 2.576 means that 99% of the Gaussian distribution random numbers are involved in the standard deviations of ± 2.576 [53]. Fig. 9 presents a comparison of heat flux predictions made with different temperature error inputs $\gamma\%$ of 1.0%, 5.0%, and 10.0%. The results of these comparisons demonstrate inconsistencies between the predicted and actual values across different temperature error inputs.

**Fig. 9.** Comparisons of incident heat fluxes inversed with different temperature error inputs.

For temperature error inputs of 1.0%, 5.0%, and 10.0%, the maximum deviations of heat flux are $1.6 \times 10^4 \text{ W/m}^2$, $2.4 \times 10^4 \text{ W/m}^2$, and $4.0 \times 10^4 \text{ W/m}^2$ respectively, with corresponding relative errors of 4.0%, 6.0%, and 10.0%. A positive correlation can be observed between the relative error of heat flux and the temperature error inputs. Furthermore, it should be noted that the precision of the MPGA-BPNN algorithm affect the accuracy of the resulting heat flux values.

3.3. Comparison of different algorithms

This section presents a comparative analysis of the numerical results generated by BPNN, GA-BPNN, and MPGA-BPNN algorithms, utilizing the Modified Flattened Gaussian Model as detailed in Table 3. Fig. 10 illustrates the comparison between the heat flux values derived from numerical simulations and those predicted by the three neural network algorithms. It is evident from Fig. 10 that the heat flux values computed by the MPGA-BPNN algorithm align more closely with the actual values. In scenarios involving larger heat fluxes, the MPGA-BPNN algorithm demonstrates superior accuracy compared to both the BPNN and GA-BPNN algorithms. For example, when the true heat flux value is $4 \times 10^5 \text{ W/m}^2$, the relative errors for BPNN, GA-BPNN, and MPGA-BPNN are 6.5%, 5.0%, and 1.0%, respectively. However, for smaller heat fluxes, the values produced by the three algorithms exhibit minimal variance. Table 6 provides a comprehensive summary of the distribution parameters of heat flux obtained by each algorithm. For instance, the relative

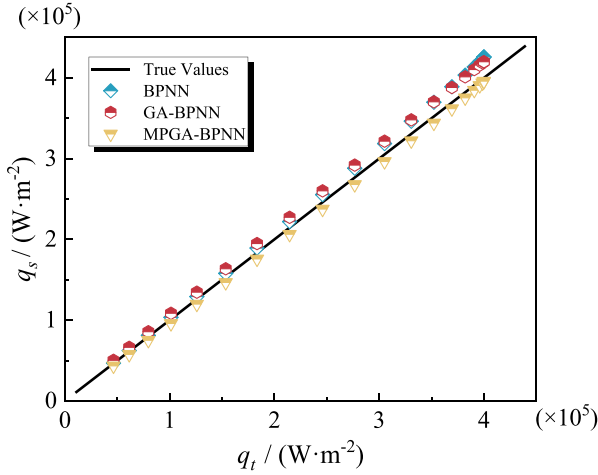


Fig. 10. Compare heat flux from numerical simulation and different algorithms.

Table 6

Distribution parameters of heat flux obtained by different algorithms.

Different algorithms	A	B	C	D
BPNN algorithm	2.83	2.74	8.7	2.13
GA-BPNN algorithm	2.86	2.85	8.84	2.1
MPGA-BPNN algorithm	2.91	2.83	8.81	1.98
True Value	3.00	3.00	9.00	2.00

errors of four distribution parameters calculated by the MPGA-BPNN algorithm are 3.0%, 5.7%, 2.1%, and 1.0%, respectively.

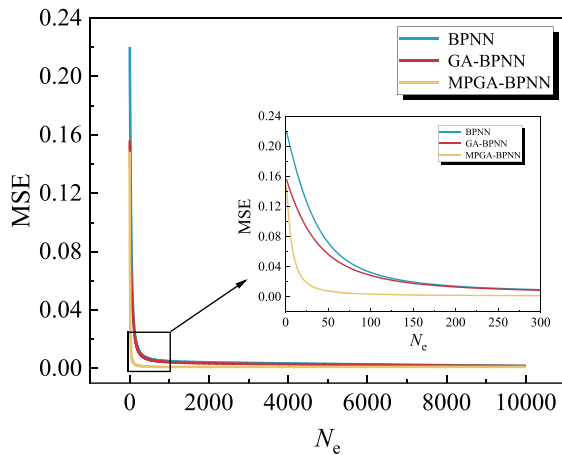
Fig. 11 shows the variations in Mean Squared Error (MSE) across different algorithms as a function of the number of epochs N_e and time cost t . As depicted in Fig. 11(a), an increase in the number of epochs results in the loss function of the network converging to its minimum value. The MSE values displayed in the figure exhibit a downward trend, indicating that the model's training process remains stable and uninterrupted. The lowest MSE values for BPNN, GA-BPNN, and MPGA-BPNN algorithms are 1.70×10^{-3} , 1.66×10^{-3} , and 1.08×10^{-3} respectively. This implies that the configurations related to input data and neural networks are appropriate, with the MPGA-BPNN algorithm demonstrating superior accuracy. After 300 epochs, the MSE value for the MPGA-BPNN algorithm has reached a state of convergence, while

the other two have not. Similarly, Fig. 11(b) illustrates the MPGA-BPNN algorithm exhibits a faster convergence rate, reaching convergence in 30 s, while the GA-BPNN and BPNN methods converge at approximately 120 s and 160 s. This indicates that the proposed MPGA-BPNN method is both efficient and faster. Moreover, the heat flux results generated by the GA-BPNN algorithm across multiple simulations are inconsistent, suggesting that the algorithm may be experiencing premature convergence towards local optimal solutions. In contrast, the MPGA-BPNN algorithm employs multiple populations to cooperatively search the solution space simultaneously, thereby considering both global and local search aspects of the algorithm. This approach effectively addresses the issue of premature convergence.

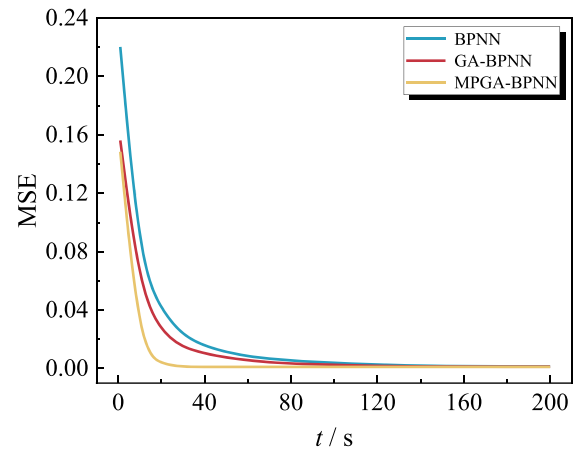
4. Experimental results and discussion

The temperature distributions on the sample front surface, as measured at various times by a thermal imager and determined by the total emissivity, are shown in Fig. 12(a). This figure displays the variation of sample front surface temperature with respect to distance from the center over a period of 1.0–5.0 s. As the distance from the sample's center increases, the temperature initially stabilizes before gradually decreasing. Conversely, when the distance from the sample's center is fixed, an increase in time results in a gradual rise in the surface temperature of the sample. However, this upward trend gradually diminishes over time. In addition, to verify the accuracy of the temperature distribution on the material surface as measured by the infrared thermal imager, the temperature values of four sampling points at different positions on the surface were verified at different times. Fig. 12(a) also shows the comparison between the infrared thermal imager measurement values and the thermocouple measurement values obtained from multiple measurements. It can be observed that the thermocouple measurement values are in good agreement with the infrared thermal imager measurement values, with a maximum difference of less than 3.0 K. It can be concluded that the measurement results of the infrared thermal imager are reliable.

Fig. 12(b) depicts the predicted heat fluxes generated by the proposed algorithm at 1.0–5.0 s. It can be observed that the values of heat fluxes remain relatively constant until the radius x increases to 9.0 mm from the sample center, after which they rapidly decrease to zero. These findings suggest that the laser beam shaper is capable of effectively transforming a Gaussian beam into a flat-top beam with a thin edge of approximately 1.0 mm in width. Furthermore, when the measurement radius x remains constant, the heat flux exhibits a gradual and gentle decline over time. Table 7 shows the distribution parameters of heat flux



(a) Number of epochs



(b) Time cost

Fig. 11. Variation of MSE of different algorithms with number of epochs and time cost.

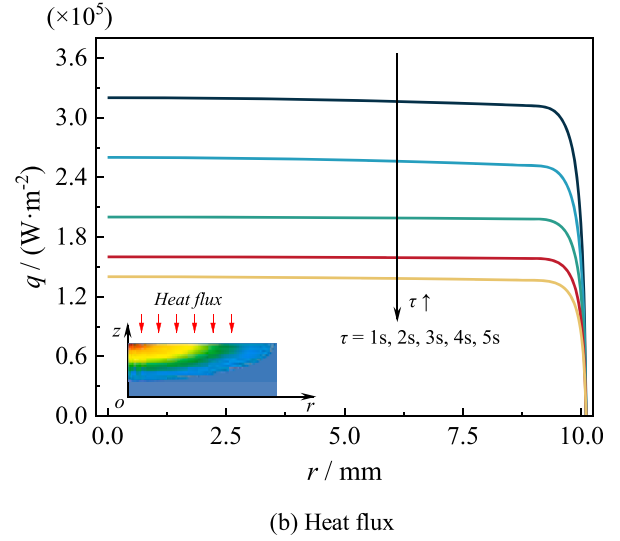
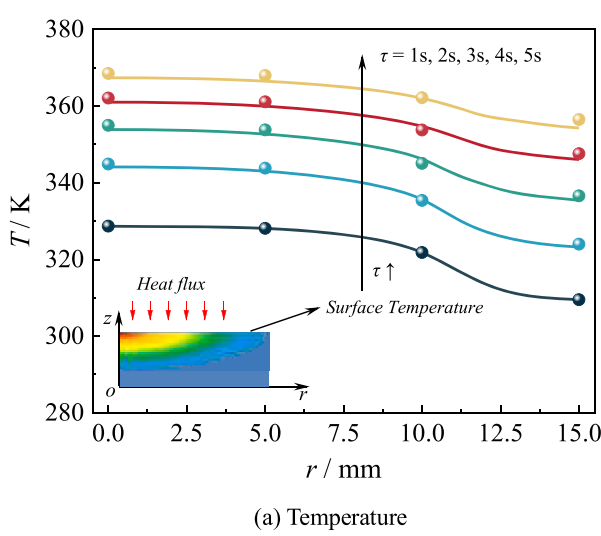


Fig. 12. Surface temperature and heat flux varied with the distance from center at different times.

at different times, as determined by MPGA-BPNN algorithms. The values of parameters A , B , and C remain constant over time. The values of A and B are relatively small and close to zero, while C fluctuates around 15.0. This indicates that the distribution shape of the laser heat flux remains unchanged over time. The parameter D demonstrates a gradual decline over time, indicative of a reduction in heat flux. An asymptotic model is employed to fit the relationship between parameter D and time τ , as shown in Eq. (18), resulting in a determination coefficient of 0.99403. The incident heat flux distribution on the sample surface can be expressed by Eq. (19).

$$D = 0.283 + 1.791 \cdot 0.743^\tau, 0 \leq \tau \leq 5 \quad (18)$$

$$q(r, \tau) = 2.0 \times 10^5 \cdot \exp\left(-0.106 \times \left(\frac{r}{15.002}\right)^2\right) \cdot \sum_{k=0}^N \frac{1}{k!} \left(0.054 \times \left(\frac{r}{15.002}\right)^2\right)^k \cdot (0.283 + 1.791 \cdot 0.743^\tau), \quad (19)$$

$$0 \leq r \leq 10, 0 \leq \tau \leq 5, N = 2$$

In addition, Table 7 shows the uniform heat fluxes at different times, as measured by a heat flux meter in our laboratory. Fig. 13 presents a comparison between the experimental heat fluxes and the calculated heat fluxes for different times. The heat fluxes at three positions on the sample surface are compared with the experimental values, and the absolute relative deviations are also given, with the maximum deviation being within 2.5%. It can be concluded that the calculated results are in good agreement with the experimental results, and that the algorithm is available for calculating the unknown heat flux of the laser.

5. Conclusions

This study presents a transient heat flux measurement method, based on a Back Propagation Neural Network (BPNN) improved by Multiple

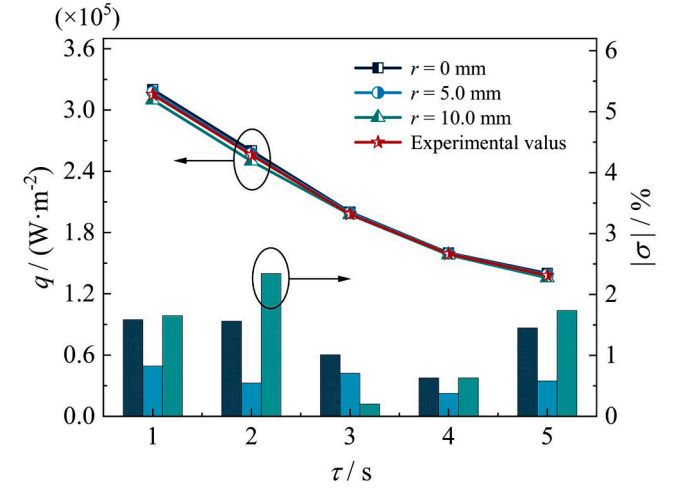


Fig. 13. Heat fluxes comparison between experimental values and calculated values at different times.

Population Genetic Algorithm (MPGA). This proposed method combines the advantages of BPNN and MPGA, while overcoming the shortcomings of GA-BPNN's reliance on initial population selection and operator parameter settings. The method incorporates both global and local search algorithms through population co-evolution with varying control parameters. The applicability of this method was evaluated through simulation experiments, and its performance in measuring laser heat flux under real conditions was also tested. The results of simulation experiments demonstrated that the computational accuracy and solution speed of this method were superior to those of BPNN and GA-BPNN. When the true heat flux value is $4 \times 10^5 \text{ W/m}^2$, the relative errors for the BPNN, GA-BPNN, and MPGA-BPNN algorithms are 6.5%, 5.0%, and 1.0% respectively. After 300 epochs, the Mean Squared Error (MSE) value for the MPGA-BPNN algorithm reached a stable point, while the other two did not, indicating a faster convergence rate for the MPGA-BPNN algorithm. The reliability of this method was confirmed by establishing several heat flux distribution models, with the maximum deviation of heat flux being less than $1.0 \times 10^4 \text{ W/m}^2$. Furthermore, the impact of network parameters and temperature input error on heat flux prediction results was analyzed. It was observed that under conditions of 12 populations and 20 neurons in the hidden layer, the relative error can be minimized without necessitating an increase in population numbers.

Table 7

Distribution parameters and experimental results of incident heat flux at different times.

Different times	A	B	C	D	$q_{\text{exp}}/\text{W}\cdot\text{m}^{-2}$
1.0s	0.10	0.07	15.02	1.60	3.15×10^5
2.0s	0.11	0.02	15.08	1.31	2.56×10^5
3.0s	0.07	0.04	14.93	1.00	1.98×10^5
4.0s	0.12	0.09	15.16	0.80	1.59×10^5
5.0s	0.13	0.05	14.82	0.71	1.38×10^5

to avoid overloading neural networks. For temperature error inputs of 1.0%, 5.0%, and 10.0%, the relative errors of heat flux were 4.0%, 6.0%, and 10.0% respectively. In terms of experimental measurements, heat flux distribution predictions at different times were conducted based on temperature measurements, providing a distribution function of heat flux over time and space at the material surface.

The current study makes a significant contribution to ongoing research on heat flux prediction and offers a viable method for obtaining time-dependent heat flux distributions at varying times. In subsequent research, the aim is to broaden the scope of the training set to encompass a wider range of heat flux distributions. Additionally, we plan to refine the algorithm to further reduce its measurement error.

CRediT authorship contribution statement

Jia-Qi Li: Writing – original draft, Methodology, Data curation, Conceptualization. **Xin-Lin Xia:** Resources, Funding acquisition. **Chuang Sun:** Investigation, Formal analysis. **Xue Chen:** Validation, Funding acquisition.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work.

Data availability

No data was used for the research described in the article.

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